

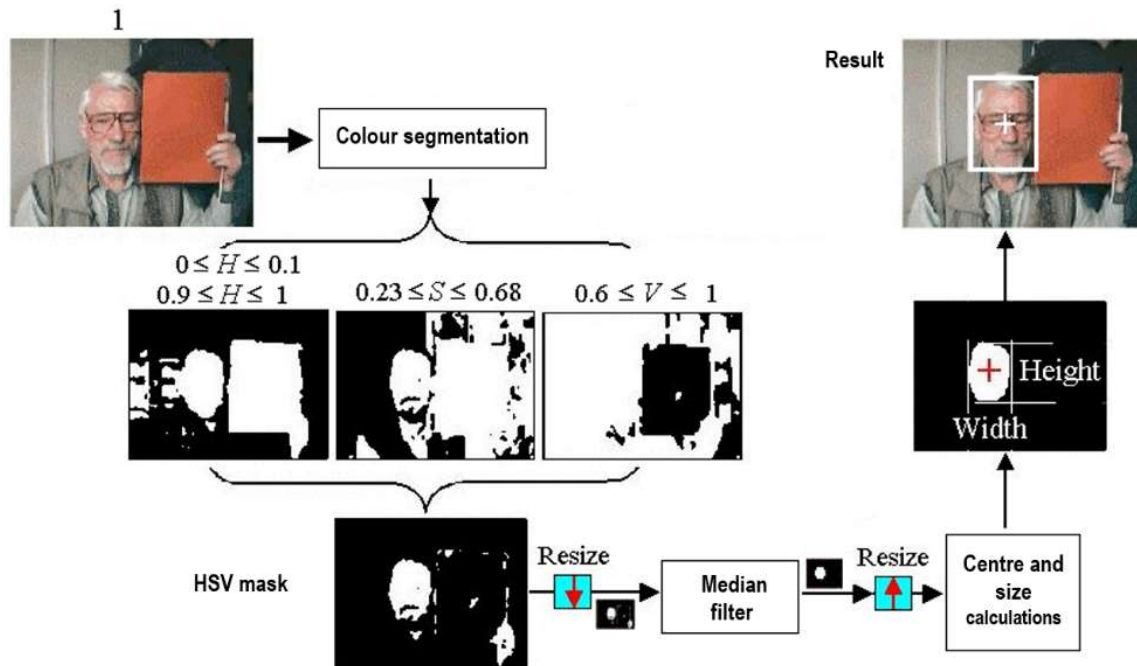
Human-Computer Interaction

Camera mouse (2): color marker detection

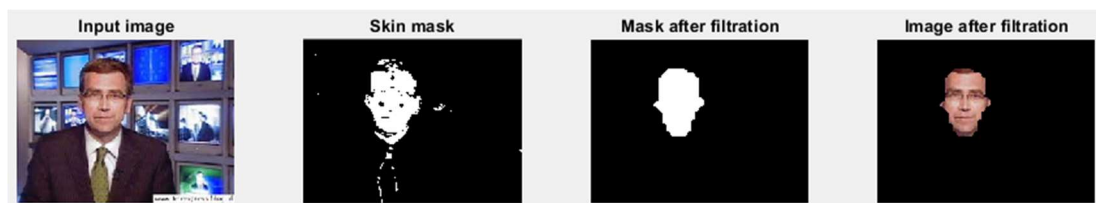
- The aim of the exercise is to implement color detection algorithm
- Our detection algorithm is based on the skin colour technique
- It can be used to detect head or hand whose movement in the next classes will be mapped to the change in the mouse pointer position on the screen
- Recommended development environments: Matlab or Python

Exercise 1 – Simple colour detector

- Write a script that creates a mask, verifying pixels against the human skin model for the HSV space:
 - $$\begin{cases} H \leq 0.1 \text{ OR } H \geq 0.9 \\ 0.2 \leq S \leq 0.6 \\ V \geq 0.4 \end{cases}$$
 - assuming the values in each of the channels H, S and V are normalized $\langle 0;1 \rangle$
 - the conversion of the color space from RGB to HSV in Matlab can be performed using the `rgb2hsv` function; an image can be read using the `imread` function
- Filter the resulting mask using scaling down and scaling back up with optional median filtering. Below is a pictorial figure (note that other threshold values have been adopted for skin color model).
 - Scale with the nearest neighbor method (Matlab function `imresize`, with the 'nearest' parameter)
 - The median filter in Matlab: `medfilt2` function
 - Use scale factors between 10% - 20% of the original size of the image



- Mask the input color image using the mask obtained



Exercise 2 - Evaluation

- Present the operation of the developed algorithm for several examples containing faces and hands
- Discuss the pros and cons of the method
- Is the skin color method sufficient and effective?
- In case both head and hand are visible in the image, can it be determined which of them will be used for steering. How can the problem be solved?

Report

- This is the second part and will be continued.
- Add conclusions from your experiments to the report.
- Don't send for now.